W1 · Machine Learning and Linear Regression

Supervised learning · Regression · Cost functions · Gradient descent

Name _____ Date _____

1. Concept Check

Choose the single best answer for each question.

1. A model sees many pairs of house sizes and sale prices, then predicts the price of a new house. What kind of learning is this?

(1) Supervised learning
(2) Grouping houses without target labels
(3) A calculation that uses no data
(4) Choosing prices at random

2. Which is a regression problem?

(1) Classifying email as spam or legitimate
(2) Classifying animals in photos as cats or dogs
(3) Predicting tomorrow's temperature as a number
(4) Classifying reviews as positive or negative

3. You have customer data without labels and want to group similar customers. Which approach is most appropriate?

(1) Assigning the same target label to every customer
(2) Finding groups with unsupervised learning
(3) Using customer IDs directly as purchase amounts
(4) Fixing the groups before looking at the data

4. In house-price prediction, x is house size and y is the actual price. Which is the input feature?

(1) The actual price
(2) The amount of prediction error
(3) The number of training iterations
(4) House size

5. In $\hat{y} = wx + b$, you keep b fixed and change only w. Which property of the line changes directly?

(1) Its slope
(2) The number of examples
(3) The actual targets
(4) The input units

1. Concept Check / Continued

Choose the single best answer for each question.

6. If $\hat{y} = 2x + 1$ and $x = 3$, what does the model predict?

(1) 3
(2) 5
(3) 7
(4) 9

7. Which best describes the purpose of a cost function?

(1) Naming the training data
(2) Summarizing the differences between predictions and actual values in one number
(3) Increasing the number of input features
(4) Always producing the correct answer

8. You compare two models on the same data using the same cost function. Model A has a lower cost than B. What does this mean?

(1) A must be more accurate on every new example
(2) A has not been trained at all
(3) B must have more data
(4) A has a smaller summary of prediction errors on this data

9. Why does gradient descent use the gradient?

(1) To adjust parameters step by step in a direction that reduces cost
(2) To change actual targets to match predictions
(3) To delete training data regardless of its usefulness
(4) To keep making the gradient larger

10. What can happen if the learning rate is too large?

(1) The model must reach the minimum in one step
(2) The number of input features automatically increases
(3) Updates can overshoot the minimum, causing the cost to increase or oscillate
(4) Every prediction becomes equal to its actual target

2. Think Like an ML Scientist

Question 11 · Read the pseudocode and find why the model does not work as intended.

You built a model that predicts house prices from size. Small parameter updates should reduce the cost, but it increases as training continues. Assume the learning rate is sufficiently small and the gradient function is implemented correctly.

```
w = 0; b = 0
alpha = 0.001
repeat 1000 times:
    dw, db = gradient(X, y, w, b)
    w = w + alpha * dw
    b = b + alpha * db
```

11(1) Identify the two lines that work against the goal of reducing cost.

11(2) Explain why the cost can increase and rewrite each incorrect line as one corrected line.

3. Open Question

Question 12 · There is no single correct answer. Describe your method and reasoning.

Suppose a tool converts sentences into lists of numbers of equal length. You use it on 100 customer reviews. How would you discover patterns among the reviews?

Describe the pattern you want to find, what you would compare, and how you would check whether it is meaningful. You do not have to assume that numerical distance necessarily measures a difference in meaning.

Equations and code are optional. You may answer with text, a diagram, or a simple flowchart.